concentration of NO_2^- in solution, which is also difficult to remove. Since ATS is less soluble than HADS, it would be beneficial to operate the scrubber at high bisulfite concentrations, to aid in the conversion of HADS to ATS and ADS. Lowering the temperature of the scrubbing liquor would also assist in the removal of the nitrogen–sulfur compounds. Optimal conditions for removal of the nitrogen–sulfur compounds could be determined by modeling the chemistry of the scrubbing liquor (Littlejohn and Chang, 1991). After separation from the scrubbing liquor, the precipitated nitrogen–sulfur compounds could be given thermal treatment to generate potassium sulfate. This could then be used to increase the potassium concentration of the scrubbing liquor rich in nitrogen–sulfur compounds to precipitate its potassium salts.

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Registry No. ATS, 63504-30-3; ADS, 14696-74-3; HADS, 21049-67-2; ADS^{3-}, 56491-90-8; HAMS, 13768-26-8.

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Vapor–Liquid Equilibria by UNIFAC Group Contribution. 5. Revision and Extension

Revised UNIFAC interaction parameters are presented for 74 pairs of groups. Parameters have been determined for 100 pairs of groups where parameters were not given in the previously available parameter table. Finally, six new main groups have been added, such that UNIFAC has been extended to cover mixtures with chlorofluorohydrocarbons, amides, glycol ethers, sulfides, morpholine, and thiophenes. A total of 223 pairs of interaction parameters have been revised or added to the parameter table.

Introduction

The UNIFAC group contribution method (Fredenslund et al., 1975, 1977) has proven to be a fast and reliable tool for predicting liquid-phase activity coefficients. The success of group contribution methods such as UNIFAC depends on the availability of group volumes (R_k), the availability of group surface areas (Q_k), and of both the availability and the reliability of group-interaction parameters (a_{mn} and a_{nm}).

An extensive table of UNIFAC group-interaction parameters was recently published by Fredenslund et al. (1977). This parameter table has been revised and extended in a series of articles in the present Journal. This article is the fifth in the series.

The purposes of the present revision and extension are as follows:

To revise the group-interaction parameters already existing in the previously published parameter table (Tiegs et al., 1987), where new experimental information warrants such revision;

To fill in gaps in the parameter table of Tiegs et al. (1987);

To add new groups so that UNIFAC may be extended to cover mixtures with chlorofluorohydrocarbons (Freons) and mixtures of interest in biotechnology (amides, ethoxyalcohols, sulfides, morpholine, and thiophenes);

To present the complete revised an updated UNIFAC parameter tables.

Parameter Estimation

The group volume and group surface parameters are based on the work of Bondi (see Fredenslund et al. (1977)) and are listed in Table I (see the paragraph at the end of this paper regarding the availability of Supplementary Material). The group-interaction parameters are calculated using the more than 14000 vapor–liquid equilibrium data sets stored in the Dortmund Data Bank (Gmehling and Onken, 1977). The data bank is continuously updated at Universität Oldenburg, Germany.

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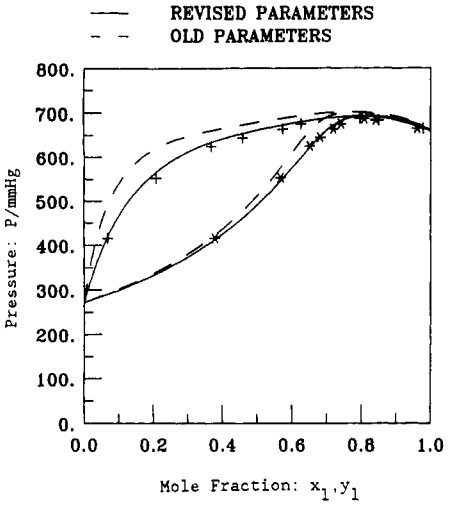


Figure 1. Vapor–liquid equilibrium data and UNIFAC predictions for ethyl formate (1)–cyclohexane (2) at 323.15 K.

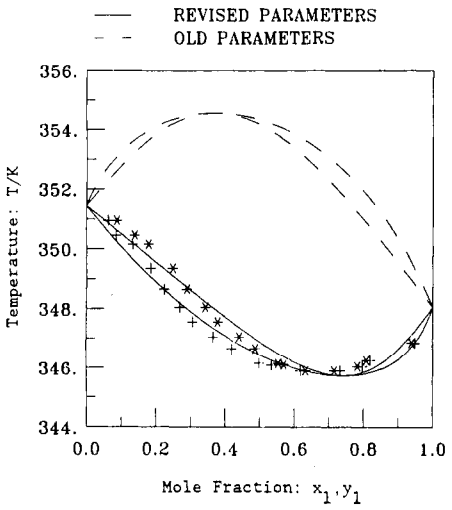


Figure 2. Vapor–liquid equilibrium data and UNIFAC predictions for butyraldehyde (1)–ethanol (2) at 1 atm.

The group-interaction parameters presented in this work are estimated by minimization

$$F_{mm} = \sum[(y_1(\text{calc}) - y_1(\text{exp}))/\sigma_y]^2 + [(P(\text{calc}) - P(\text{exp}))/\sigma_p]^2 \quad (1)$$

where y_1 is the vapor-phase mole fraction of component 1 and P is the pressure. Based on experience, the various σ_y and σ_p were fixed at the values 0.003 and 0.5 mmHg, respectively. The Levenberg–Marquardt method was used for the minimization.

Parameter Revisions

On the basis of new, additional information in the Dortmund Data Bank, it has been deemed worthwhile to revise the parameters for some group interactions. A total of 74 pairs of group-interaction parameters have been reestimated. The parameters that have been changed are indicated by asterisks in Table II (see supplementary Material). Examples of improvements in the predictions of vapor–liquid equilibria using UNIFAC are shown in Figures 1 and 2. (Note that the system shown in Figure 2 is a very narrowly boiling mixture.)

Parameters Not Previously Available

New experimental and reliable vapor–liquid equilibrium data stored in the Dortmund Data Bank have made it possible to estimate interaction parameters that were previously not available. One hundred pairs of groups have been added to the parameter table. This is indicated with a circle (O) in Table II. Figure 3 shows an example of a prediction which has become possible with the present revision and extension.

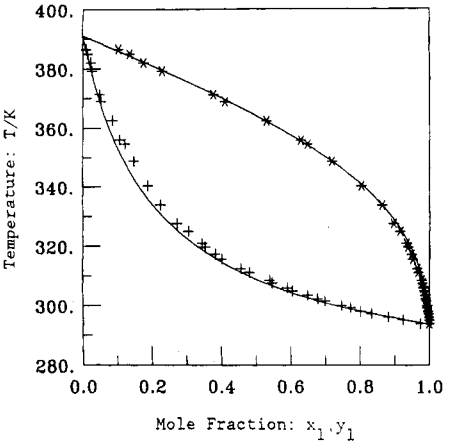


Figure 3. Vapor–liquid equilibrium data and UNIFAC predictions for acetaldehyde (1)–acetic acid (2) at 1 atm.

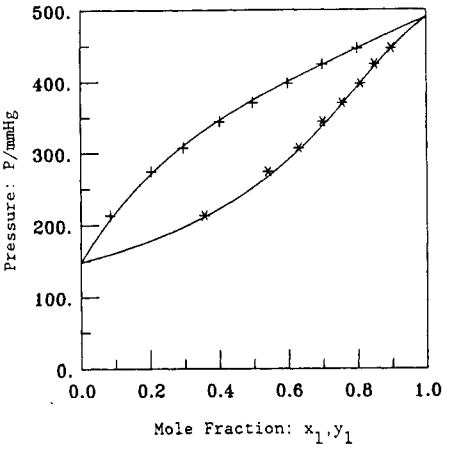


Figure 4. Vapor–liquid equilibrium data and UNIFAC predictions for 1,1,2-trichloro-1,2,2-trifluoroethane (1)–benzene (2) at 308.15 K.

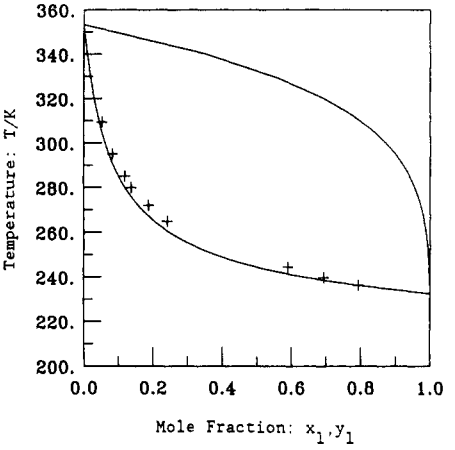


Figure 5. Vapor–liquid equilibrium data and UNIFAC predictions for difluorochloromethane (1)–benzene (2) at 1 atm.

New Groups and Parameters

Freon substitution and biotechnology are of increasing importance in chemical engineering. For this reason the

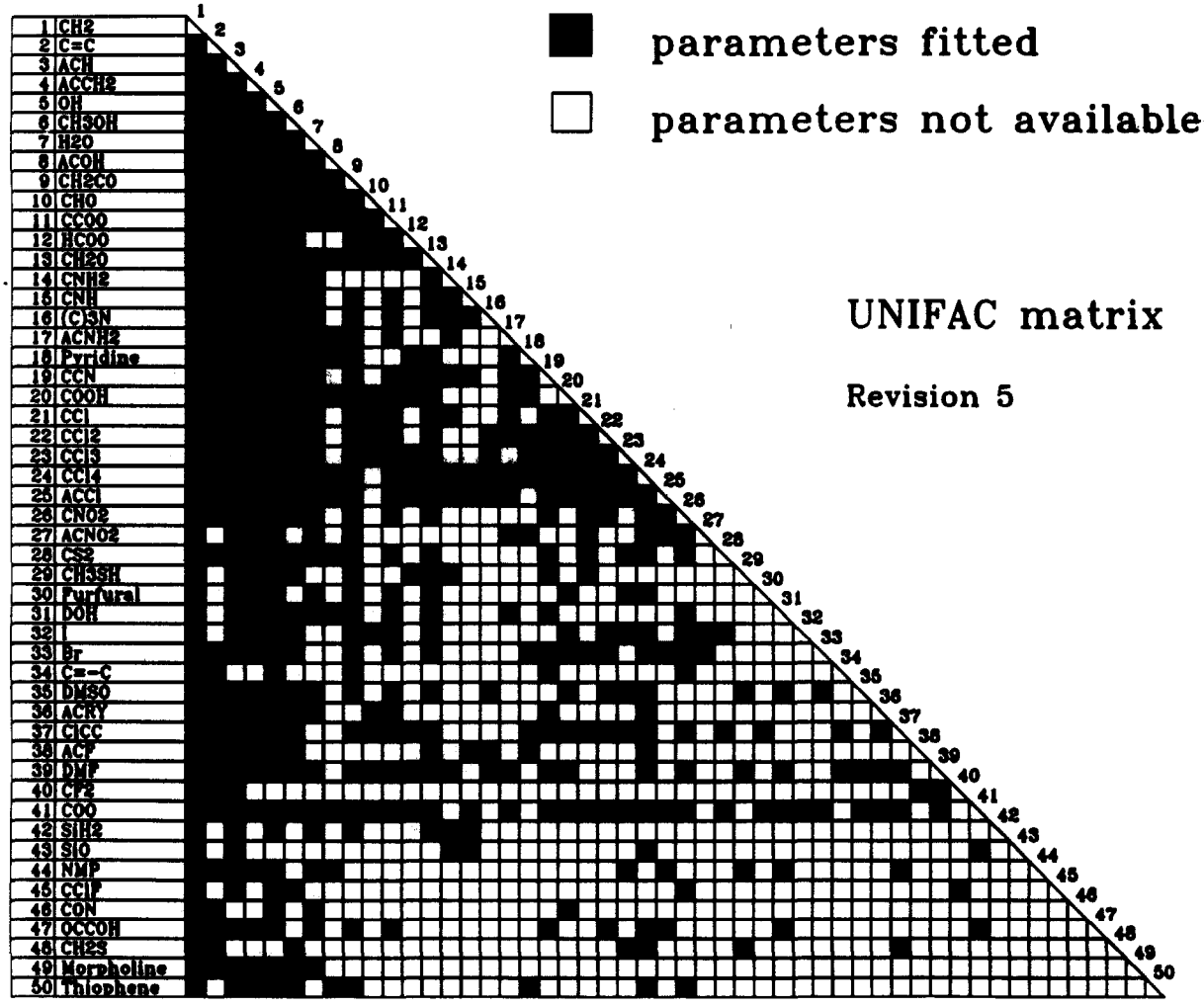


Figure 6. UNIFAC group-interaction parameter matrix. Revision 5.

following groups have been added:

Main Group 45 "CClF". This group may be used to represent Freons. There are eight Freon subgroups. Perhaps surprisingly, the same group-interaction parameters may be used to represent phase equilibria of mixtures with Freons, irrespective of how many chlorine and/or fluorine atoms are present on the carbon atom. Examples of such predictions are shown on Figures 4 and 5.

Main Group 46 "CON". The group "CON" may be used to represent amides. This group has six subgroups.

Main Group 47 "OCCOH". It has not previously been possible to represent glycol ethers in UNIFAC due to proximity effects between the ether group and the alcohol group. A new group including both of these has therefore been defined. There are two subgroups.

Main Group 48 "CH_2S". Three subgroups are available enabling sulfides to be included in UNIFAC.

Main Group 49 "Morpholine". Due to its heterocyclic structure with both an O and NH in the ring, morpholine is defined as one main group.

Main Group 50 "Thiophene". Thiophene is one separate group. Two additional subgroups within this main group allow representatives of mono- and disubstituted thiophenes.

Parameters for New Groups

Volume and surface area parameters for the subgroups covered by the new main groups (45–50) were determined by using the compilation by Bondi (1968). They are listed toward the end of Table I.

Interaction parameters involving the new main groups have been determined for 49 pairs of groups. They are indicated with a # in Table II.

Results

The availability of UNIFAC group-interaction parameters is shown in Figure 6, where dark areas indicate where parameters are available and light areas indicate where, due to lack of reliable data, parameters are not available. Table I gives group volume and surface area values for all the groups presently included in UNIFAC. Table II lists all the presently available group-interaction parameters.

Conclusion

The revised parameters of Table II have improved the reliability of UNIFAC vapor–liquid equilibrium predictions, and the additional parameters in Tables I and II have extended the range of applicability of the method.

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Registry No. Morphine, 110-91-8.

Supplementary Material Available: Table I listing UNIFAC group volume (R_k) and surface area (Q_k) parameters and Table II listing UNIFAC group interaction parameters

for prediction of vapor–liquid equilibria at 250–425 K (16 pages). Ordering information is given on any current masthead page.

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Decomposition of Dichlorodifluoromethane on BPO_4 Catalyst

Catalytic decomposition of dichlorodifluoromethane (Freon 12) was carried out on BPO_4 catalyst, and the durability of this catalyst against inorganic fluorines, the reaction products, was investigated. The decomposition of Freon 12 proceeded on the acid sites of the BPO_4, although their acid strength was not very large. Although the BPO_4 catalyst had the highest durability among the catalysts ever examined, it was also deactivated during prolonged use. The analysis by ESCA (electron spectroscopy for chemical analysis) indicated that inorganic fluorines attacked the boron to produce volatile boron fluorides, which were eliminated from this catalyst during the reaction. CaO absorbed inorganic chlorines and fluorines. Thus the efficiency of the BPO_4-catalyzed decomposition of Freon 12 was improved by the combined use of CaO which protected the BPO_4 catalyst from poisoning by inorganic fluorines.

There is a great concern about the depletion of the ozone layer caused by chlorofluorocarbons (CFC's). Although their production will be diminished in the future owing to the United Nations Environmental Protection Protocol for CFC regulation adopted in Montreal, Canada, in 1987, some of them are still being used now, and the preserved CFC's will inevitably continue to increase. Thus an efficient method to destruct CFC's must be urgently developed.

CFC's can be destructed by various methods: reduction by use of sodium naphthalenide (Oku et al., 1988, 1989), laser-induced decomposition (Hudgens, 1977; Zitter et al., 1990), catalytic hydrogenolysis (Witt et al., 1981), and catalytic decomposition (Colussi and Amorebieta, 1987). Of these methods, catalytic decomposition seems most practical. Chlorotrifluoromethane and 1,1,2-trichloro-1,2,2-trifluoroethane were readily decomposed over Fe_2O_3 supported on activated charcoal above 450 °C in the presence of water vapor (Okazaki and Kurosaki, 1989). Gold supported on Al_2O_3 showed high activity in decomposing dichlorodifluoromethane (Aida et al., 1990). Hydrogen fluoride and molecular fluorine were produced during the decomposition of these CFC's. Although these inorganic fluorines are highly reactive and cause the deactivation of the catalysts, there have been scarce data on the durability of the catalysts used for the decomposition of CFC's.

Previously we reported that multiple-step catalytic combustion is effective to dispose of chlorinated organic compounds (Imamura et al., 1989a,b). This process is composed of the following three steps. The first step is a catalytic decomposition of chlorinated organics into inorganic chlorines and a carbonaceous moiety (mainly to CO and CO_2). The second step is the absorption of inorganic chlorines by basic oxides such as CaO to protect the combustion catalysts to be employed in the last step where complete combustion of the carbonaceous moiety is accomplished. This process could be applied to the combustion of dichlorodifluoromethane (Freon 12), and the active catalysts in the first step were acidic oxides such as zeolite, mordenite, silica–alumina, and titania–silica. Although these acid catalysts, especially titania–silica, were durable against inorganic chlorines, they were deactivated rapidly during the decomposition of Freon 12 due to an attack by inorganic fluorines (Imamura et al., 1990). We, therefore, investigated the activity and durability of various catalysts including acid catalysts other than those mentioned above, precious-metal catalysts, transition-metal oxides, and so forth (Imamura et al., 1991). The examination of these catalysts led us to conclude that it was impossible to design robust catalysts against fluorines by using metals or silicon as starting elements. As a nonmetal acid catalyst, supported sulfuric acid was found to be active. However, sulfur escaped as sulfur oxides during the reaction. Finally, it was found that a phosphate and boron-based compound, BPO_4, was the only promising catalyst. The present paper deals with the catalytic performance of the BPO_4 catalyst in the decomposition of Freon 12. The recovery of fluorines and chlorines by CaO and its effect to protect the BPO_4 catalyst from attack by inorganic fluorines are also described.

Experimental Section

Materials. The BPO_4 was prepared by the reaction of boric acid with phosphoric acid and was calcined at 550 °C in air for 3 h (Ozaki, 1980). Its structure was determined as BPO_4 with an X-ray diffraction analyzer, and its BET surface area (S_w) was 12.6 m^2/g. Its particle size as determined by a SEM technique was about 0.6 μm for the primary particles and about 100 μm for the secondary particles, respectively. The primary particles aggregated to form large secondary particles. CeO_2 (S_w 80.7 m^2/g) was prepared by precipitation from aqueous cerium(III) nitrate, followed by calcination at 550 °C in air for 3 h (Imamura

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